Mathematical Induction II

This Lecture

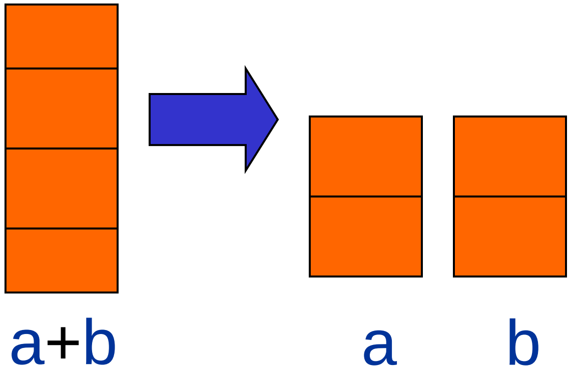
We will continue our discussions on mathematical induction.

The new elements in this lecture are a few variants of induction:

- Strong induction
- Well Ordering Principle
- Invariant Method

Unstacking Game

- Start: a stack of boxes
- Move: split any stack into two stacks of sizes a,b>0
- Scoring: ab points
- Keep moving: until stuck
- Overall score: sum of move scores



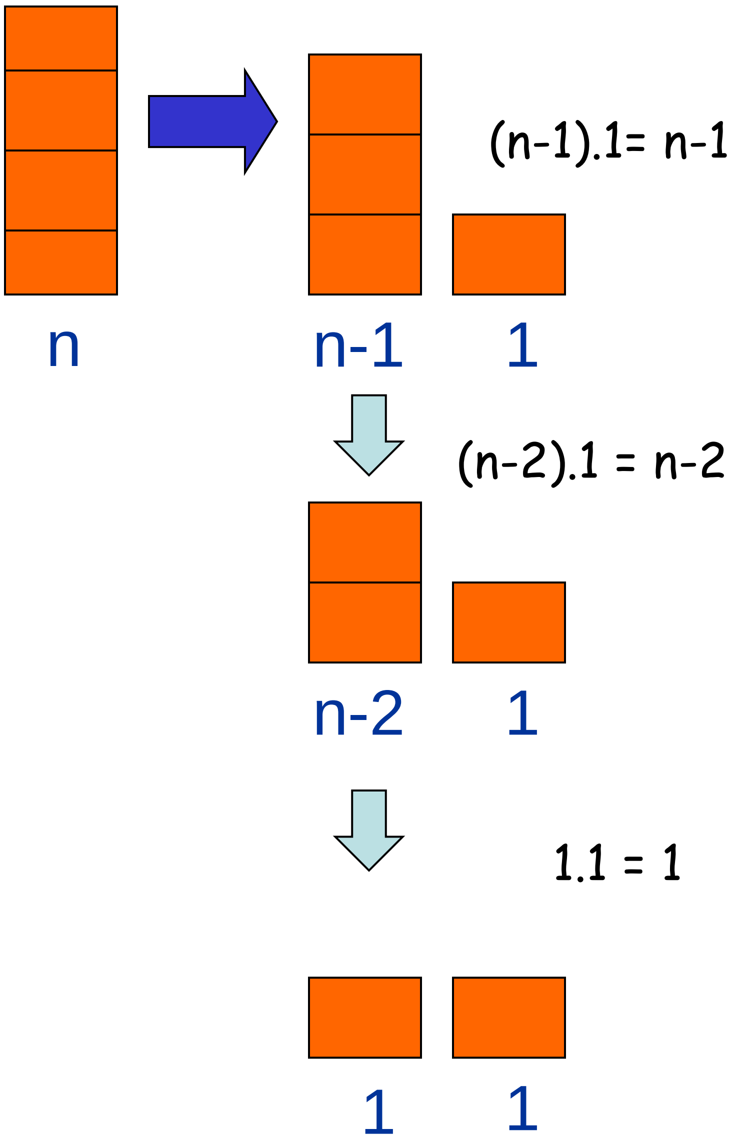
Unstacking Game

What is the best way to play this game?

Suppose there are n boxes.

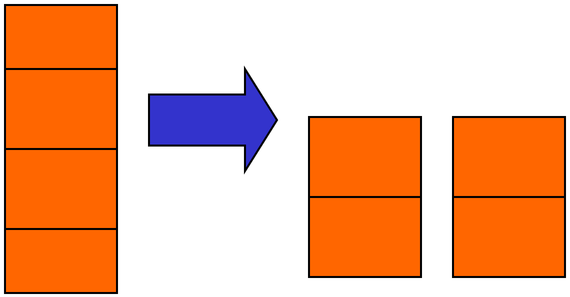
What is the score if we just take the box one at a time?

$$\sum_{i=0}^{n-1} i = \frac{n(n-1)}{2}$$



Unstacking Game

What is the best way to play this game?



Suppose there are n boxes.

What is the score if we cut the stack into half each time?

Say n=8, then the score is $\underbrace{1 \times 4 \times 4}_{\text{first round}} + \underbrace{2 \times 2 \times 2}_{\text{second}} + \underbrace{4 \times 1}_{\text{third}} = 28$

Say n=16, then the score is 8x8 + 2x28 = 120

Not better than the first strategy!

$$\frac{n(n-1)}{2}$$

Unstacking Game

Claim: Every way of unstacking gives the same score.

Claim: Starting with size n stack, final score will be $\frac{n(n-1)}{2}$

Proof: by Induction with *Claim*(n) as hypothesis

Base case n = 0:

$$\text{score} = 0 = \frac{0(0-1)}{2}$$

Claim(0) is okay.

Unstacking Game

Inductive step. assume for n-stack,

and then prove $C(n+1)$:

$(n+1)$-stack score = $$\frac{(n+1)n}{2}$$

Case $n+1 = 1$. verify for 1-stack:

score = 0 $= \frac{1(1-1)}{2}$ Put n+1 = 1 and hence n = (1-1) in above

$C(1)$ is okay.

Unstacking Game

Case $n+1 > 1$. So split into an a-stack and b-stack,

where $a + b = n + 1$.

$$\boxed{(a + b)\text{-stack score} = ab + a\text{-stack score} + b\text{-stack score}}$$

by induction:

$$a\text{-stack score} = \frac{a(a-1)}{2}$$

$$b\text{-stack score} = \frac{b(b-1)}{2}$$

Unstacking Game

(a + b)-stack score = ab + a-stack score + b-stack score

$$ab + \frac{a(a-1)}{2} + \frac{b(b-1)}{2} =$$

$$\frac{2ab + a^2 - a + b^2 - b}{2} = \frac{(a+b)^2 - (a+b)}{2} =$$

$$\frac{(a+b)((a+b)-1)}{2} = \frac{(n+1)n}{2}$$

so $C(n+1)$ is okay. We're done!

Induction Hypothesis

Wait: we *assumed* $C(a)$ and $C(b)$ where $1 \leq a, b \leq n$.

But by induction can only assume $C(n)$

the fix: revise the induction hypothesis to

$$Q(n) ::= \forall m \leq n.\, C(m)$$

In words, it says that we assume the claim is true for all numbers up to n.

Proof goes through fine using $Q(n)$ instead of $C(n)$.

So it's OK to assume $C(m)$ for all $m \leq n$ to prove $C(n+1)$.

Strong Induction

Strong induction

Prove $P(0)$.

Then prove $P(n+1)$ assuming *all* of

$P(0), P(1), \ldots, P(n)$ (instead of just $P(n)$).

Conclude $\forall n.P(n)$

equivalent

Ordinary induction

$0 \to 1, 1 \to 2, 2 \to 3, \ldots, n\text{-}1 \to n.$

So by the time we got to $n+1$, already know *all* of

$P(0), P(1), \ldots, P(n)$

The point is: assuming P(0), P(1), up to P(n), it is often easier to prove P(n+1).

Divisibility by a Prime

Theorem. Any integer $n > 1$ is divisible by a prime number.

Remember this slide?

Now we can prove it

by strong induction

very easily. In fact

we can prove a

stronger theorem

very easily.

- Let n be an integer.
- If n is a prime number, then we are done.
- Otherwise, $n = ab$, both are smaller than n.
- If a or b is a prime number, then we are done.
- Otherwise, $a = cd$, both are smaller than a.
- If c or d is a prime number, then we are done.
- Otherwise, repeat this argument, since the numbers are getting smaller and smaller, this will eventually stop and we have found a prime factor of n.

Idea of induction.

Prime Products

Theorem. Any integer $n > 1$ is divisible by a prime number.

Theorem: Every integer > 1 is a product of primes.

Proof: (by strong induction)

- Base case is easy.
- Suppose the claim is true for all $2 <= i < n$.
- Consider an integer n.
- If n is prime, then we are done.
- So $n = k \cdot m$ for integers k, m where $n > k,m > 1$.
- Since k,m smaller than n,
- *By the induction hypothesis, both k and m are product of primes*

$$k = p_1 \cdot p_2 \cdot \cdot \cdot p_{94}$$

$$m = q_1 \cdot q_2 \cdot \cdot \cdot q_{214}$$

Prime Products

Theorem: Every integer > 1 is a product of primes.

…So

$$n = k \cdot m = p_1 \cdot p_2 \cdots p_{94} \cdot q_1 \cdot q_2 \cdots q_{214}$$

is a prime product.

$\therefore$ This completes the proof of the induction step.

Postage by Strong Induction

Available stamps:

5¢

3¢

What amount can you form?

Theorem: Can form any amount $\geq$ 8¢

Prove by strong induction on *n*.

$P(n)$ ::= can form n¢.

Postage by Strong Induction

Base case (n = 8):

8¢:

Inductive Step: assume m¢ for $8 \le m < n$,

then prove n¢

cases:

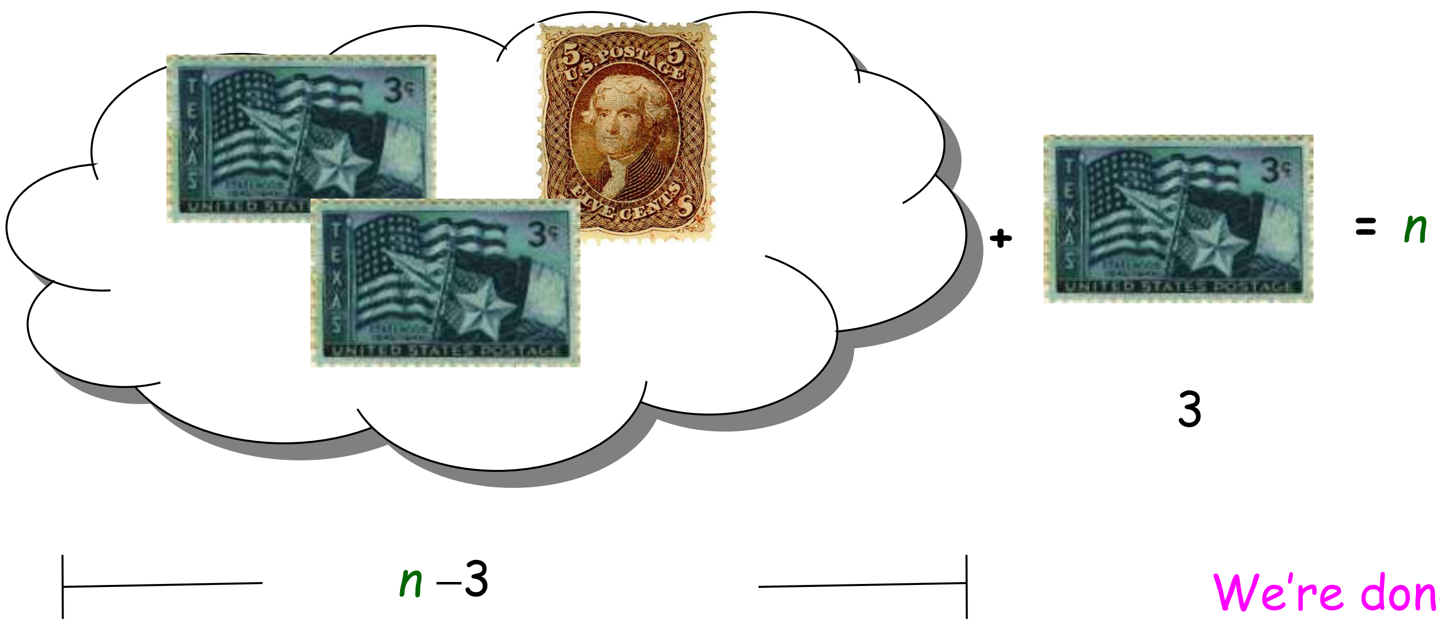
n=9:

n=10:

Postage by Strong Induction

case $n \geq 11$: let $m = n - 3$.

now $n \geq m \geq 8$, so by induction hypothesis have:



We're done!

In fact, use at most two 5-cent stamps!

	+3	+3	+3	+3	+3	+3	
8	11	14	17	20	23	26	...
9	12	15	18	21	24	27	...
10	13	16	19	22	25	28	...

In fact, use at most two 5-cent stamps!

Postage by Strong Induction

Given an unlimited supply of 5 cent and 7 cent stamps,

what postages are possible?

Theorem: For all n >= 24,

it is possible to produce n cents of postage from 5¢ and 7¢ stamps.

This Lecture

- Strong induction
- Well Ordering Principle
- Invariant Method

Well Ordering Principle

Axiom

Every nonempty set of *nonnegative integers* has a *least element.*

This is an axiom equivalent to the principle of mathematical induction.

Note that some similar looking statements are not true:

Every nonempty set of *nonnegative rationals* has a *least element.* — NO!

Every nonempty set of ~~*nonnegative*~~ *integers* has a *least element.* — NO!

Proof by Contradiction

Theorem: $\sqrt{2}$ is irrational.

Proof (by contradiction): [sketch of the proof]

- Suppose $\sqrt{2}$ was rational.
- Choose *m*, *n* integers without common prime factors (always possible)

  such that $\sqrt{2} = \dfrac{m}{n}$

- Show that *m* and *n* are both even, thus having a common factor 2,

  a **contradiction**!

Proof by Contradiction

Theorem: $\sqrt{2}$ is irrational.

$$\sqrt{2} = \frac{m}{n}$$

m, *n* integers without common prime factors (always possible)

Proof (by contradiction):

Want to prove both m and n are even.

$$\sqrt{2} = \frac{m}{n}$$

$$\sqrt{2}n = m$$

$$2n^2 = m^2$$

so *m* is even.

For an integer n, n is even if and only if n^2 is even.

so can assume $m = 2l$

$$m^2 = 4l^2$$

$$2n^2 = 4l^2$$

$$n^2 = 2l^2$$

so *n* is even.

Well Ordering Principle

Prove that $\sqrt{2}$ is irrational

Suppose, $\sqrt{2} = m/n$; m, n positive integers,

Using WOP we can ensure that n is the least among possible denominators.

Now, we have $2 = m^2/n^2$
$\Rightarrow 2n^2 = m^2$

Now LHS is even; So, RHS is also even. So, m^2 is even
Hence m is even; let $m = 2k$ for an integer k.

Then we have,
$2n^2 = m^2 = (2k)^2 = 4k^2$
$\Rightarrow n^2 = 2k^2$

RHS is even; so LHS is also even. So, n is even.
let $n = 2r$ for an integer r.

So, $\sqrt{2} = m/n = 2k/2r = k/r$

Since m and n are positive, so are k and r.

And $\boxed{r = n/2 < n}$

CONTRADCTON!

Well Ordering Principle in Proofs

To prove ``$\forall n \in \mathbb{N}.\ P(n)$'' using WOP:

1. Define the set of *counterexamples*

$$C ::= \{n \in \mathbb{N} \mid \neg P(n)\}$$

2. Assume C is not empty.
3. By WOP, have minimum element $m_0 \in C$.
4. Reach a contradiction (*somehow*) –

   usually by finding a member of C that is $< m_0$.

5. Conclude no counterexamples exist. QED

Well Ordering Principle

Prove that $\sqrt{2}$ is irrational

Suppose, $\sqrt{2} = m/n$; m, n positive integers,

Using WOP we can ensure that $\boxed{\text{n is the least among possible denominators.}}$

Steps 1-3

Now, we have 2 = m^2/n^2
⇒2n^2 = m^2

Now LHS is even; So, RHS is also even. So, m^2 is even
Hence m is even; let m = 2k for an integer k.

Then we have,
2n^2 = m^2 = (2k)^2 = 4k^2
⇒n^2 = 2k^2

RHS is even; so LHS is also even. So, n is even.
let n = 2r for an integer r.

So, $\sqrt{2}$ =m/n = 2k/2r = k/r

Since m and n are positive, so are k and r.

And $\boxed{r = n/2 < n}$

CONTRADCTON!

Steps 4

Non-Fermat Theorem

It is difficult to prove there is no positive integer solutions for

$$a^3 + b^3 = c^3$$

Fermat's theorem

But it is easy to prove there is no positive integer solutions for

$$4a^3 + 2b^3 = c^3$$

Non-Fermat's theorem

Hint: Prove by contradiction using well ordering principle…

Non-Fermat Theorem

$$4a^3 + 2b^3 = c^3$$

Suppose, by contradiction, there are integer solutions to this equation.

By the well ordering principle, there is a solution with $|a|$ smallest.

In this solution, a,b,c do not have a common factor.

Otherwise, if $a=a'k$, $b=b'k$, $c=c'k$,

then a',b',c' is another solution with $|a'| < |a|$,

contradicting the choice of a,b,c.

Claim A. There is a solution in which a,b,c do not have a common factor.

Non-Fermat Theorem

$$4a^3 + 2b^3 = c^3$$

On the other hand, we prove that every solution must have a,b,c even.

This will contradict Claim A, and complete the proof.

First, since c^3 is even, c must be even. (because odd power is odd).

a. Proposition: for an odd number m, m^i is odd for all non-negative integer i.
b. If m is an odd number, then m^i is odd for all non-negative integer i.
c. If m^i is NOT odd then m is NOT odd

Let c = 2c', then

$$4a^3 + 2b^3 = (2c')^3$$

$$4a^3 + 2b^3 = 8c'^3$$

$$\boxed{b^3 = 4c'^3 - 2a^3}$$

Non-Fermat Theorem

$$\boxed{b^3 = 4c'^3 - 2a^3}$$

Since b^3 is even, b must be even. (because odd power is odd).

Let b = 2b', then $$(2b')^3 = 4c'^3 - 2a^3$$

$$8b'^3 = 4c'^3 - 2a^3$$

$$a^3 = 2c'^3 - 4b'^3$$

Since a^3 is even, a must be even. (because odd power is odd).

$$\boxed{\text{There a,b,c are all even, contradicting Claim A}}$$

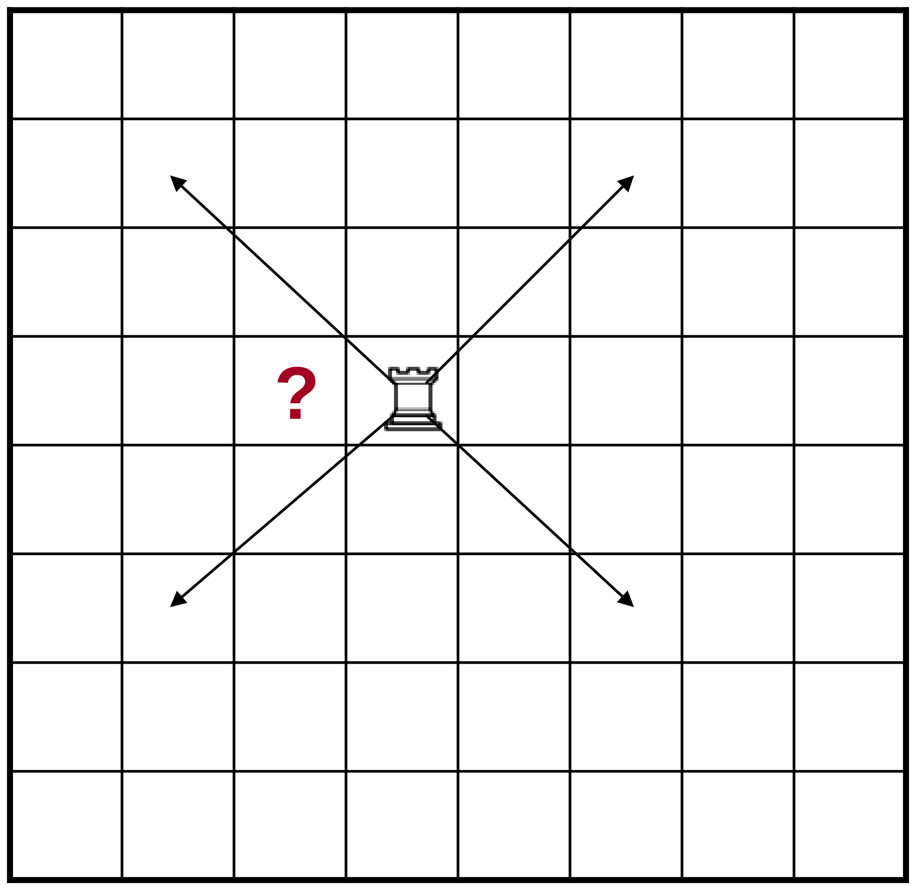
This Lecture

- Strong induction
- Well Ordering Principle
- Invariant Method

A Chessboard Problem

A rook ♖ can only move along a diagonal

Can a rook move from its current position to the question mark?



A Chessboard Problem

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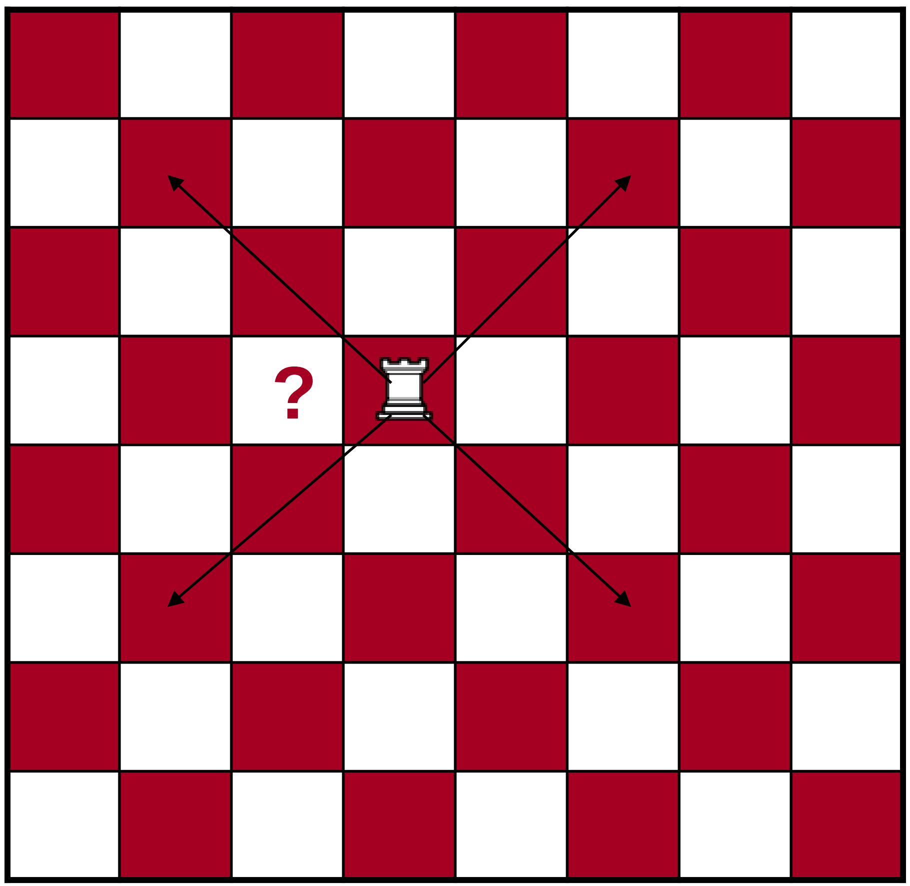
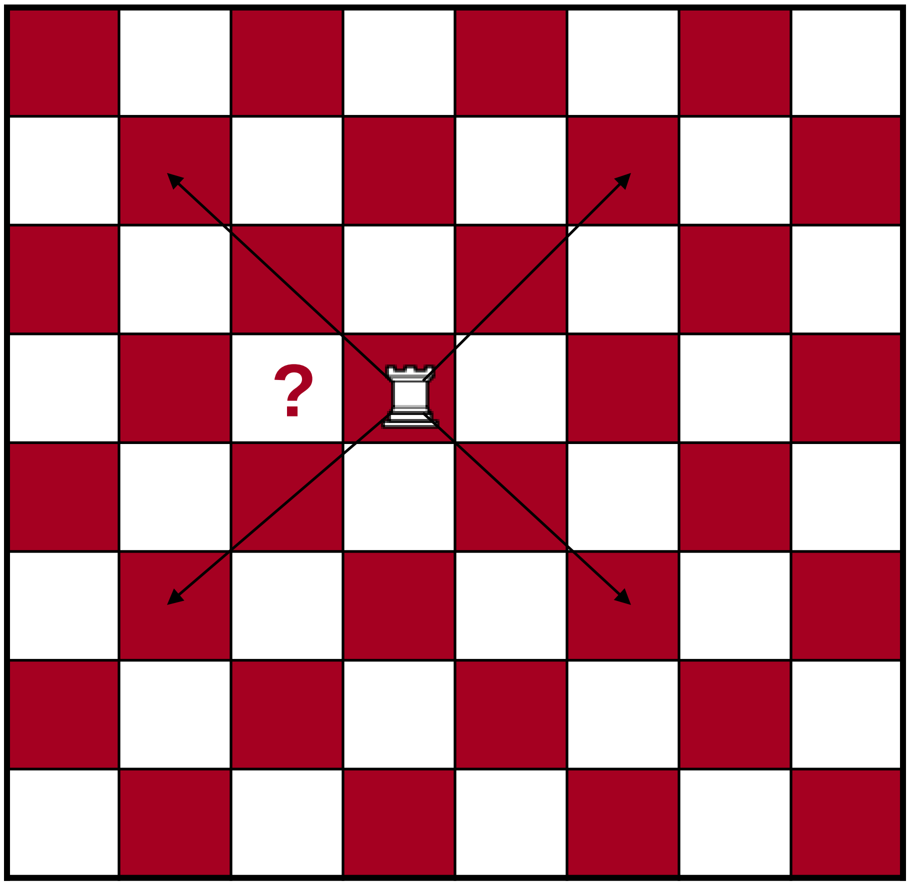


Impossible!

Why?

A Chessboard Problem

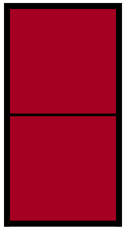


Invariant!

1. The rook is in a red position.
2. A red position can only move to a red position by diagonal moves.
3. The question mark is in a white position.
4. So it is impossible for the rook to go there.

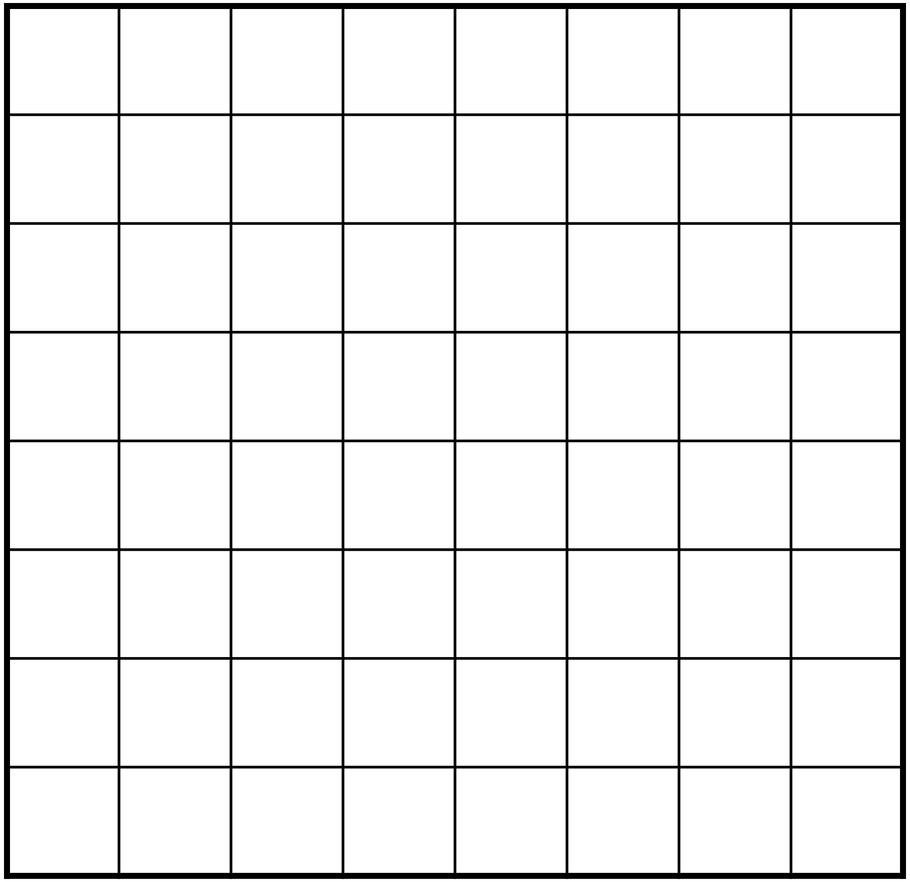
This is a simple example of the invariant method.

Domino Puzzle



An 8x8 chessboard, 32 pieces of dominos

Can we fill the chessboard?



Domino Puzzle

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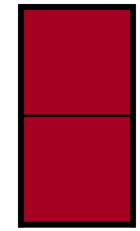
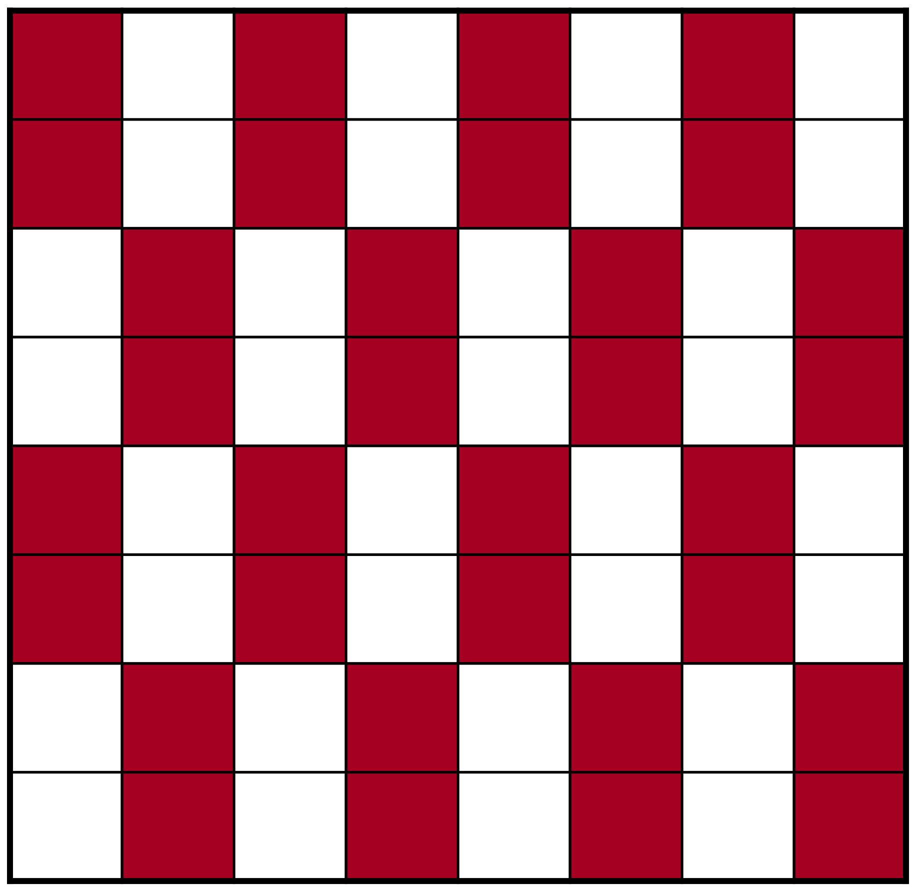
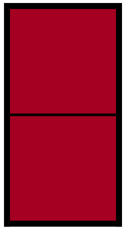




Easy!

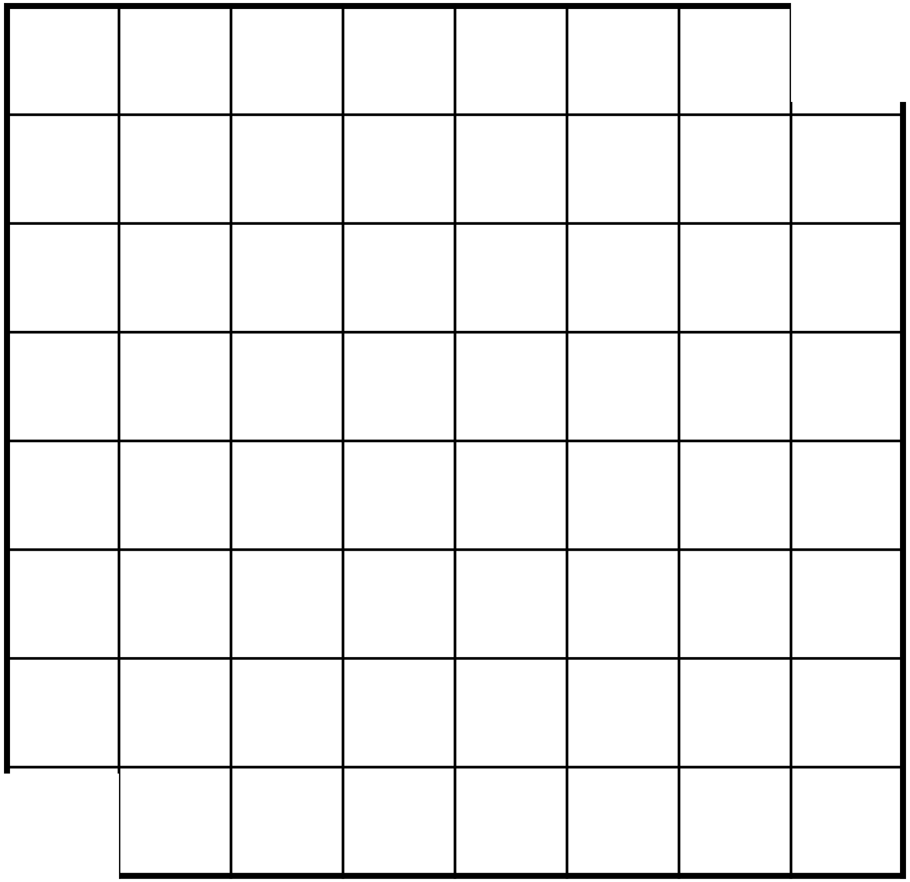
Domino Puzzle

An 8x8 chessboard with two holes, 31 pieces of dominos

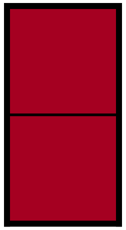


Can we fill the chessboard?

Easy??



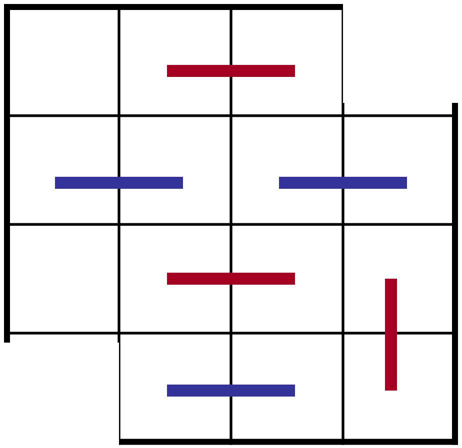
Domino Puzzle



An 4x4 chessboard with two holes, 7 pieces of dominos

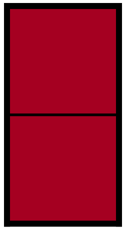
Can we fill the chessboard?

Impossible!



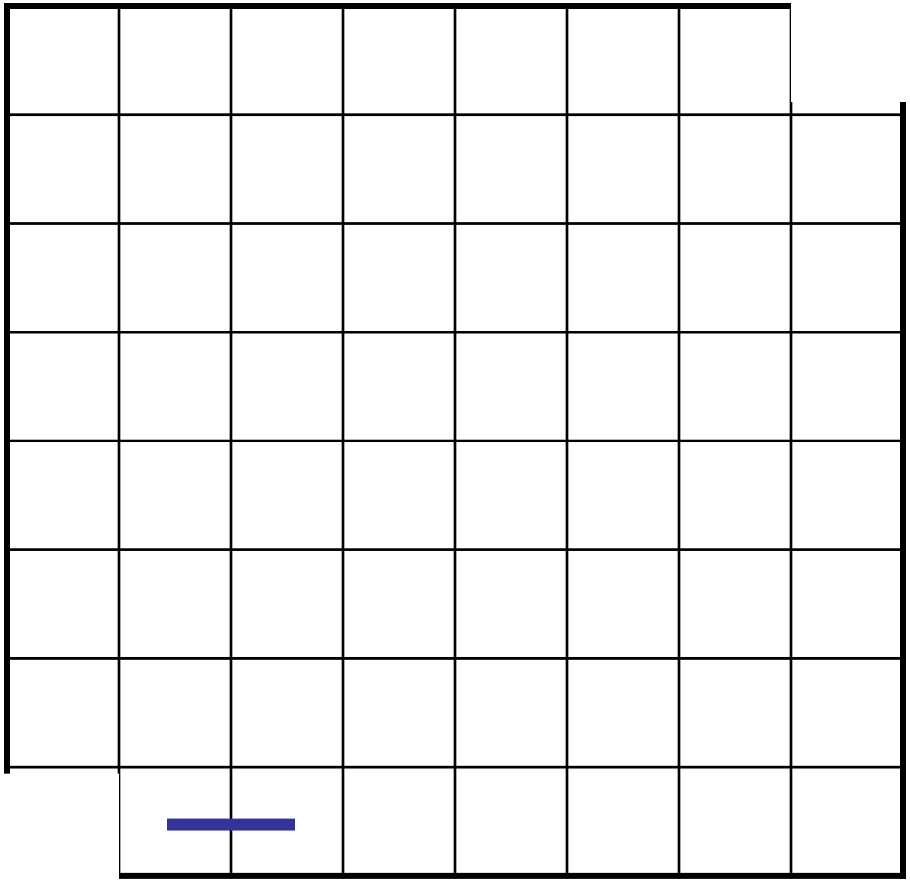
Domino Puzzle

An 8x8 chessboard with two holes, 31 pieces of dominos



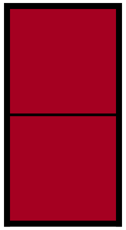
Can we fill the chessboard?

Then what??

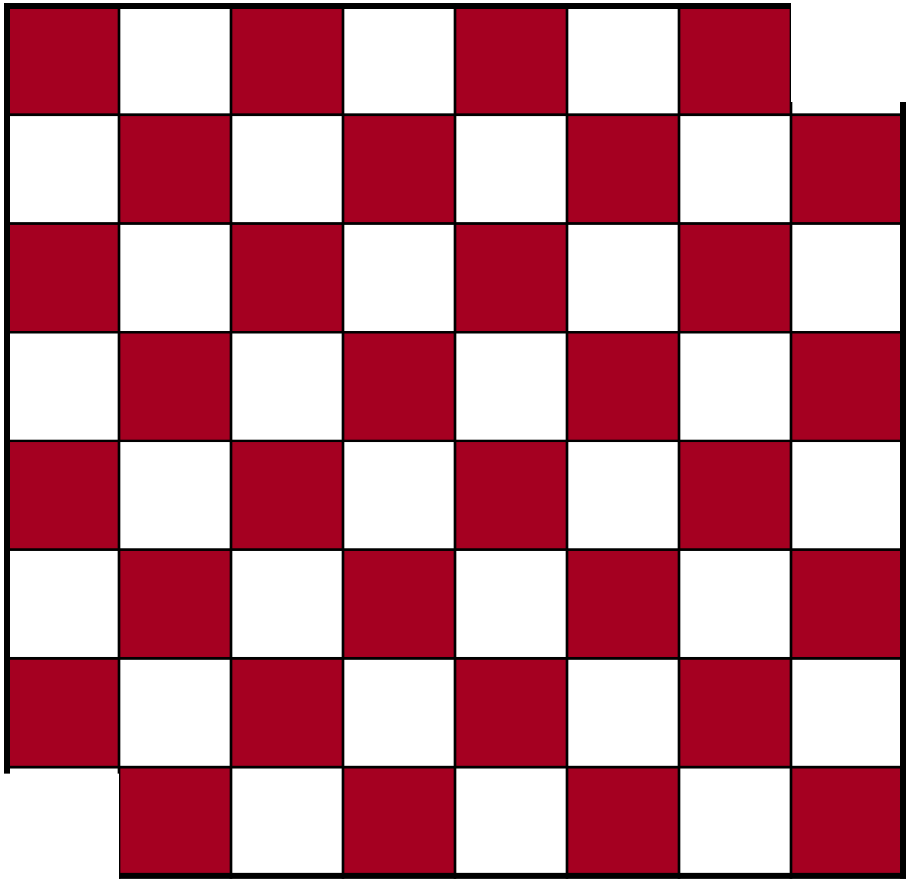


Domino Puzzle

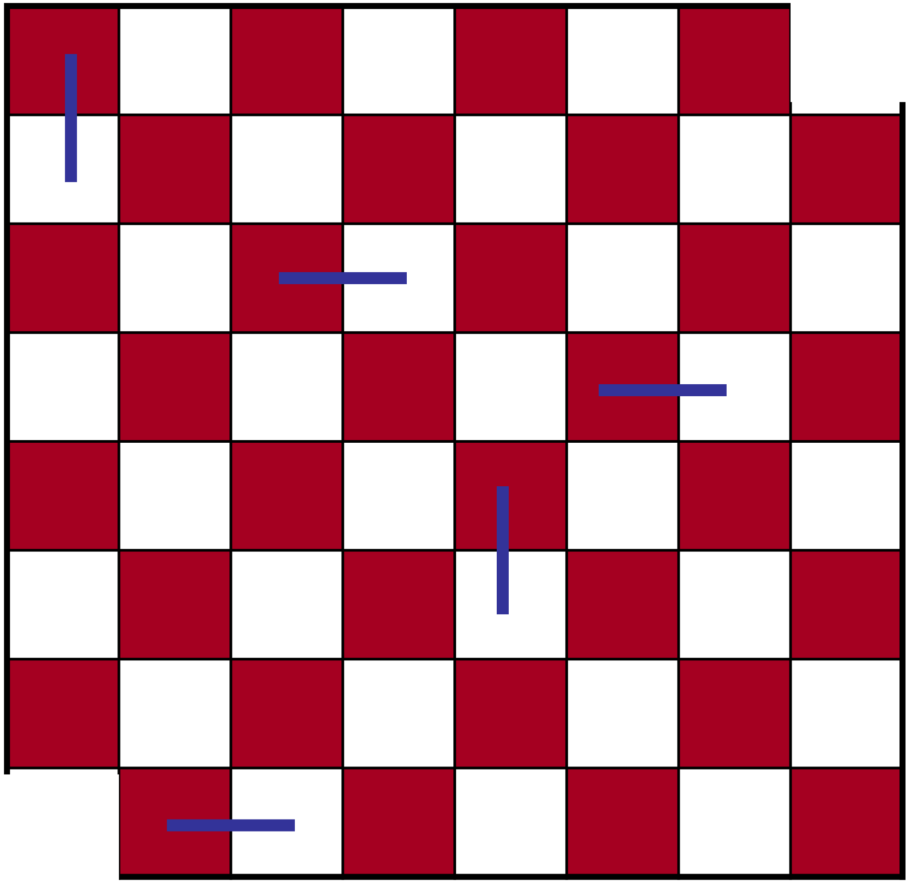
An 8x8 chessboard with two holes, 31 pieces of dominos



Can we fill the chessboard?



Domino Puzzle



Invariant!

1. Each domino will occupy one white square and one red square.

2. There are 32 blue squares but only 30 white squares.

3. So it is impossible to fill the chessboard using only 31 dominos.

This is another example of the invariant method.

Invariant Method

1. Find properties (**the invariants**) that are satisfied throughout the whole process (**by induction**).
2. Show that the target do not satisfy the properties.
3. Conclude that the target is not achievable.

In the rook example, the invariant is the colour of the position of the rook.

In the domino example, the invariant is that any placement of dominos will occupy the same number of red positions and white positions.

Very useful in analysis of algorithms.

Quick Summary

Induction is perhaps the most important proof technique in computer science.

For example it is very important in proving the correctness of an algorithm

(by invariant method) and also analyzing the running time of an algorithm.

There is no particular example that you should remember.

The point here is to understand the principle of mathematical induction

(the way that you "reduce" a large problem to smaller problems),

and apply it to the new problems that you will encounter in future!

Possibly the only way to learn this is to do more exercises!